DERIVED TENSOR FUNCTORS ARE FUNCTORIAL IN THE BIMODULE, AND QUASI-ISOMORPHISMS GIVE ISOMORPHISMS OF FUNCTORS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the property-(P) assembly of the transformation, the filtered-colimit reduction, the induction on the filtration with the split-inclusion short exact sequences, and the final graded-piece evaluation are all retained verbatim.

Editorial note. The label lemma-functoriality-bc names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Computation of derived tensor by property (P) modules. Same chapter, Lemma lemma-derived-bc. Identifies the derived tensors - tensor_A^L N and - tensor_A^L N' with plain tensors on property-(P) modules, the starting point for the transformation and its quasi-isomorphism analysis.
- Exactness of filtered colimits and the filtration of property-(P) modules. Algebra, Lemma algebra-lemma-directed-colimit-exact; Same chapter, construction of property (P). Lets the tensor with M be read through the filtered system of its filtration stages, whose split inclusions give the short exact sequences the induction runs on.

Source reference. Original lemma 00DB: sources/stacks-chow/tags.txt, source lines 491–491. Prerequisite pointer invoked at proof line(s) 5592; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09LS: sources/stacks-dga/pinned-original-source.tex, source lines 5541–5541. Prerequisite pointer invoked at proof line(s) 5581; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be a ring. Let (A, d) and (B, d) be differential graded R-algebras. Let $f : N \to N'$ be a homomorphism of differential graded (A, B)-bimodules. Then f induces a morphism of functors*

$$1 \otimes f : - \otimes_A^{\mathbf{L}} N \longrightarrow - \otimes_A^{\mathbf{L}} N'$$

If f is a quasi-isomorphism, then $1 \otimes f$ is an isomorphism of functors.

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Proof. Let M be a differential graded A-module with property (P). Then $1 \otimes f : M \otimes_A N \to M \otimes_A N'$ is a map of differential graded B-modules. Moreover, this is functorial with respect to M. Since the functors $- \otimes_A^{\mathbf{L}} N$ and $- \otimes_A^{\mathbf{L}} N'$ are computed by tensoring on objects with property (P) (Lemma 09LS) we obtain a transformation of functors as indicated.

Assume that f is a quasi-isomorphism. Let $F_\bullet$ be the given filtration on M. Observe that $M \otimes_A N = \operatorname{colim} F_i(M) \otimes_A N$ and $M \otimes_A N' = \operatorname{colim} F_i(M) \otimes_A N'$. Hence it suffices to show that $F_n(M) \otimes_A N \to F_n(M) \otimes_A N'$ is a quasi-isomorphism (filtered colimits are exact, see Algebra, Lemma 00DB). Since the inclusions $F_n(M) \to F_{n+1}(M)$ are split as maps of graded A-modules we see that

$$0 \to F_n(M) \otimes_A N \to F_{n+1}(M) \otimes_A N \to F_{n+1}(M)/F_n(M) \otimes_A N \to 0$$

is a short exact sequence of differential graded B-modules. There is a similar sequence for N' and f induces a map of short exact sequences. Hence by induction on n (starting with $n = -1$ when $F_{-1}(M) = 0$) we conclude that it suffices to show that the map $F_{n+1}(M)/F_n(M) \otimes_A N \to F_{n+1}(M)/F_n(M) \otimes_A N'$ is a quasi-isomorphism. This is true because $F_{n+1}(M)/F_n(M)$ is a direct sum of shifts of A and the result is true for $A[k]$ as $f : N \to N'$ is a quasi-isomorphism. $\square$